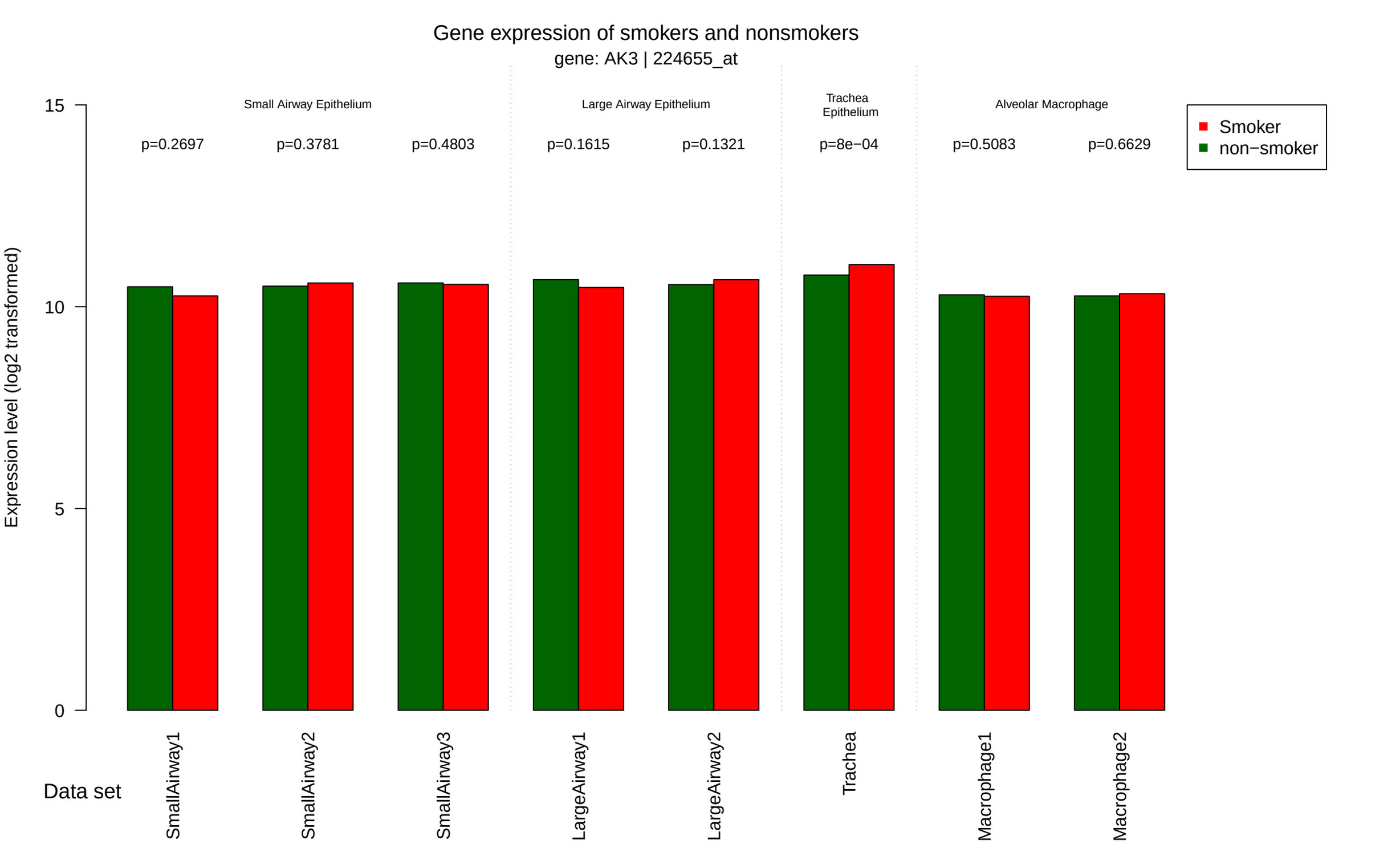
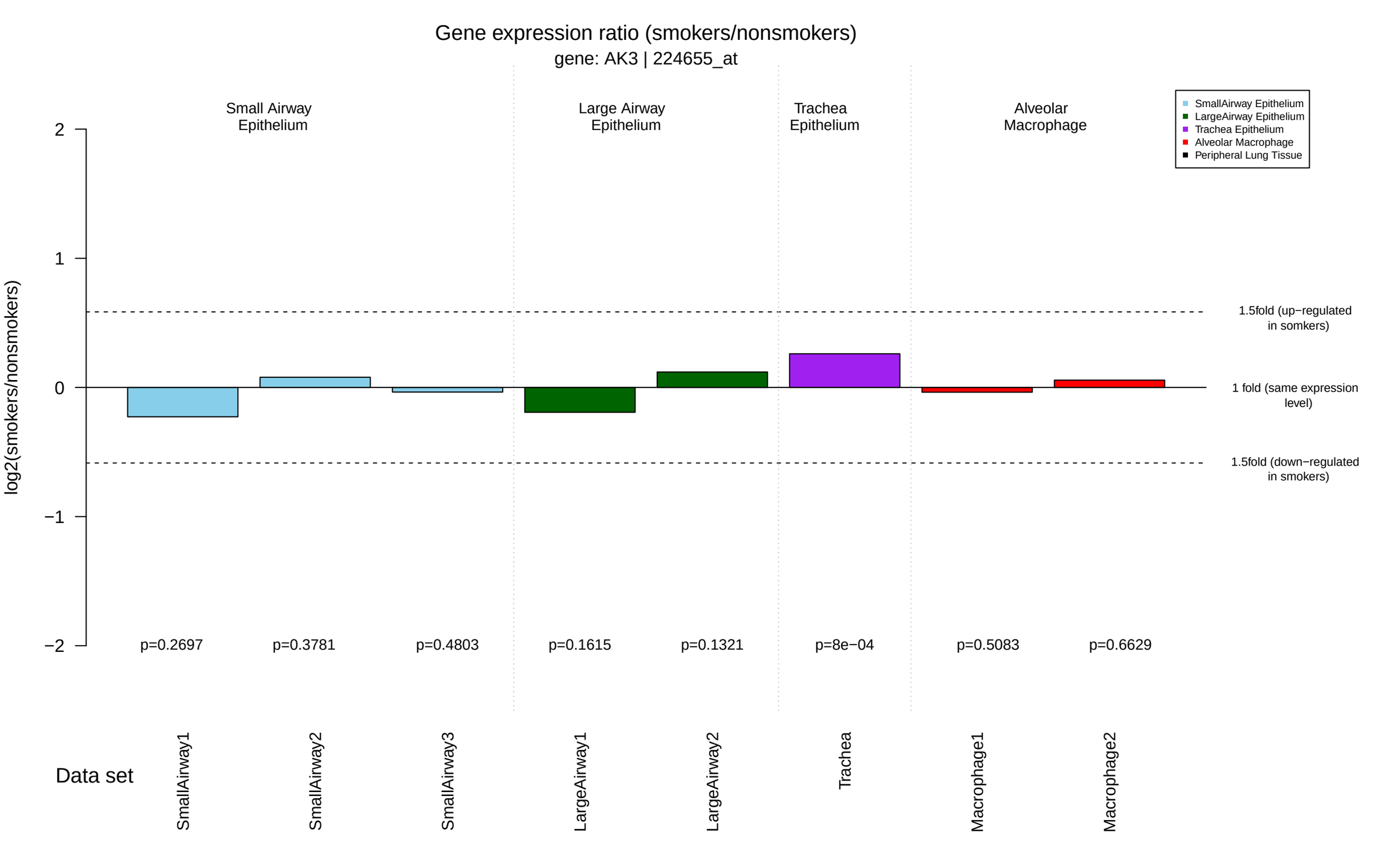
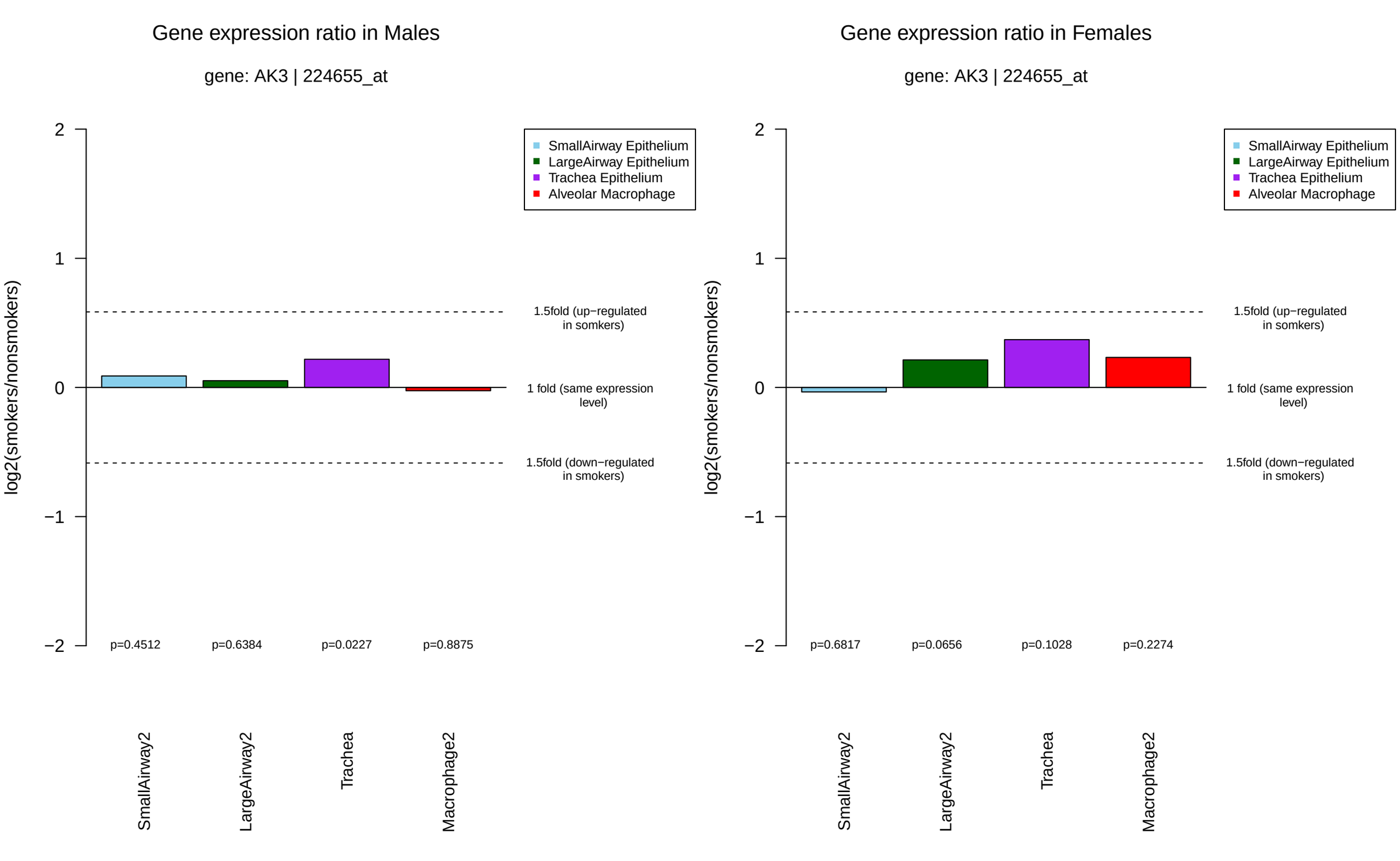
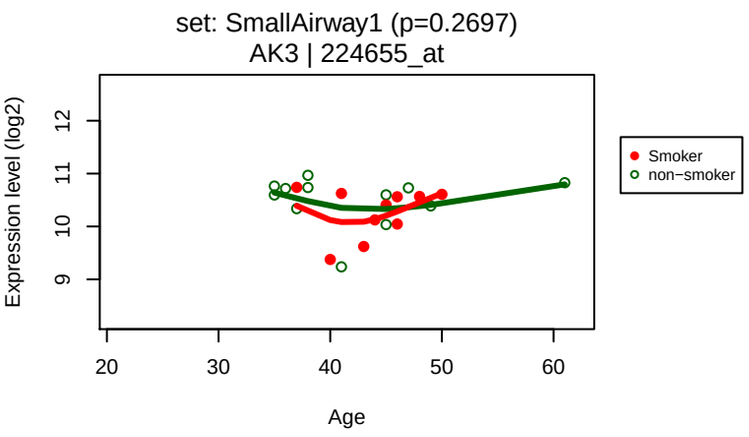
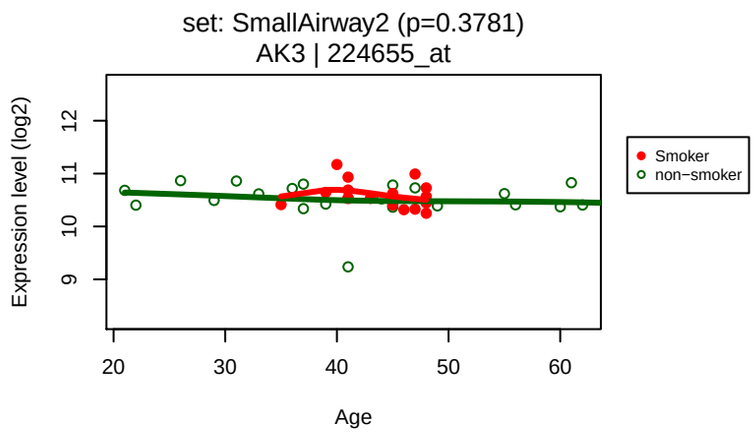
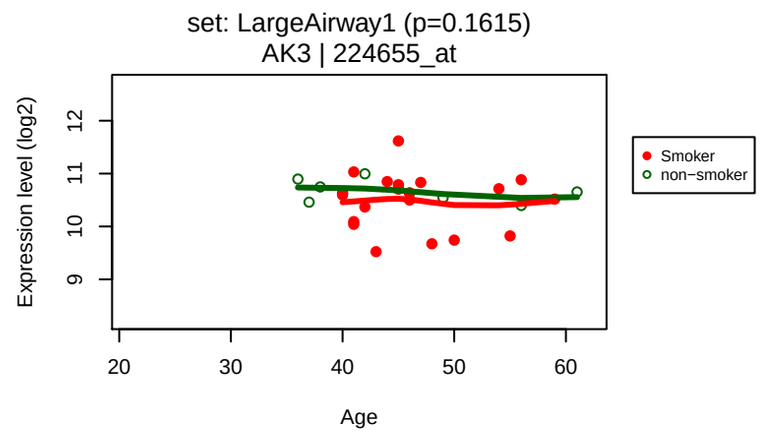

Gene expression of smokers and nonsmokers
gene: AK3 | 224655_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2697
p=0.3781
p=0.4803
p=0.1615
p=0.1321
p=8e−04
p=0.5083
p=0.6629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

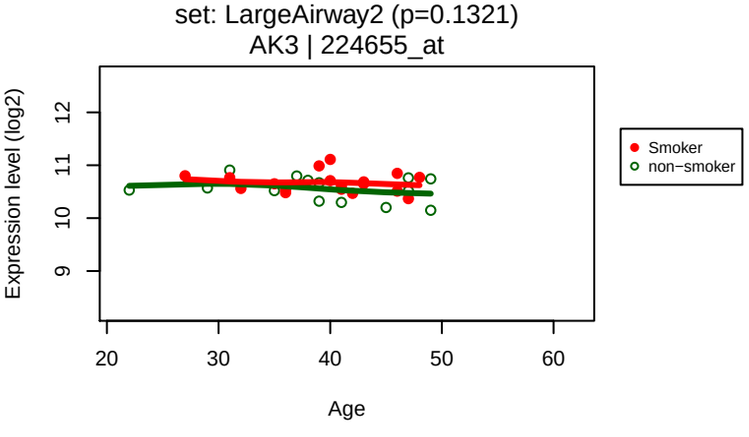

Gene expression ratio (smokers/nonsmokers)
gene: AK3 | 224655_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2697
p=0.3781
p=0.4803
p=0.1615
p=0.1321
p=8e−04
p=0.5083
p=0.6629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

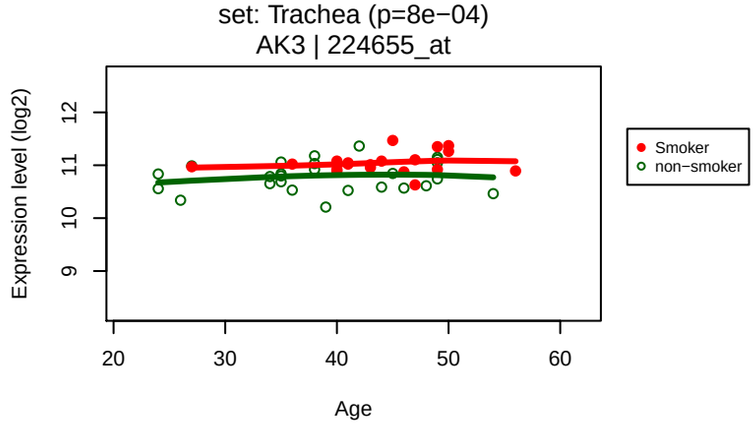

Gene expression ratio in Males
gene: AK3 | 224655_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4512
p=0.6384
p=0.0227
p=0.8875
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK3 | 224655_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6817
p=0.0656
p=0.1028
p=0.2274
SmallAirway2
LargeAirway2
Trachea
Macrophage2

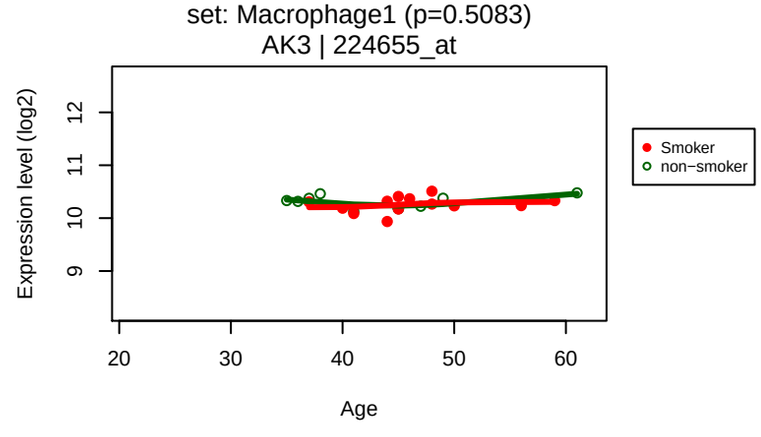

set: SmallAirway1 (p=0.2697)
AK3 | 224655_at
Expression level (log2)
Age
Smoker
non−smoker

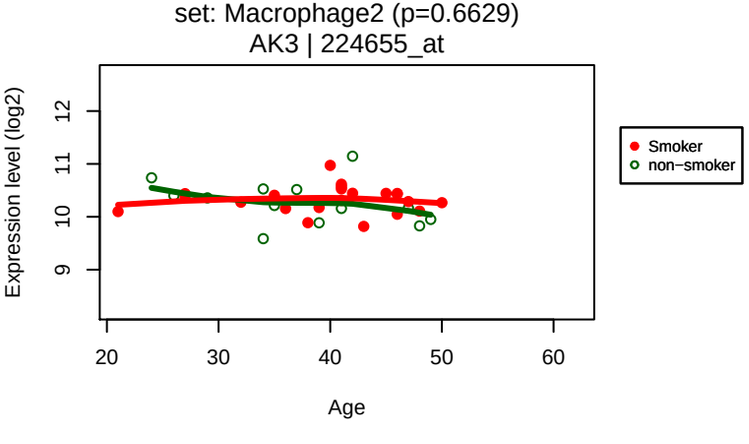

set: SmallAirway2 (p=0.3781)
AK3 | 224655_at
Expression level (log2)
Age
Smoker
non−smoker

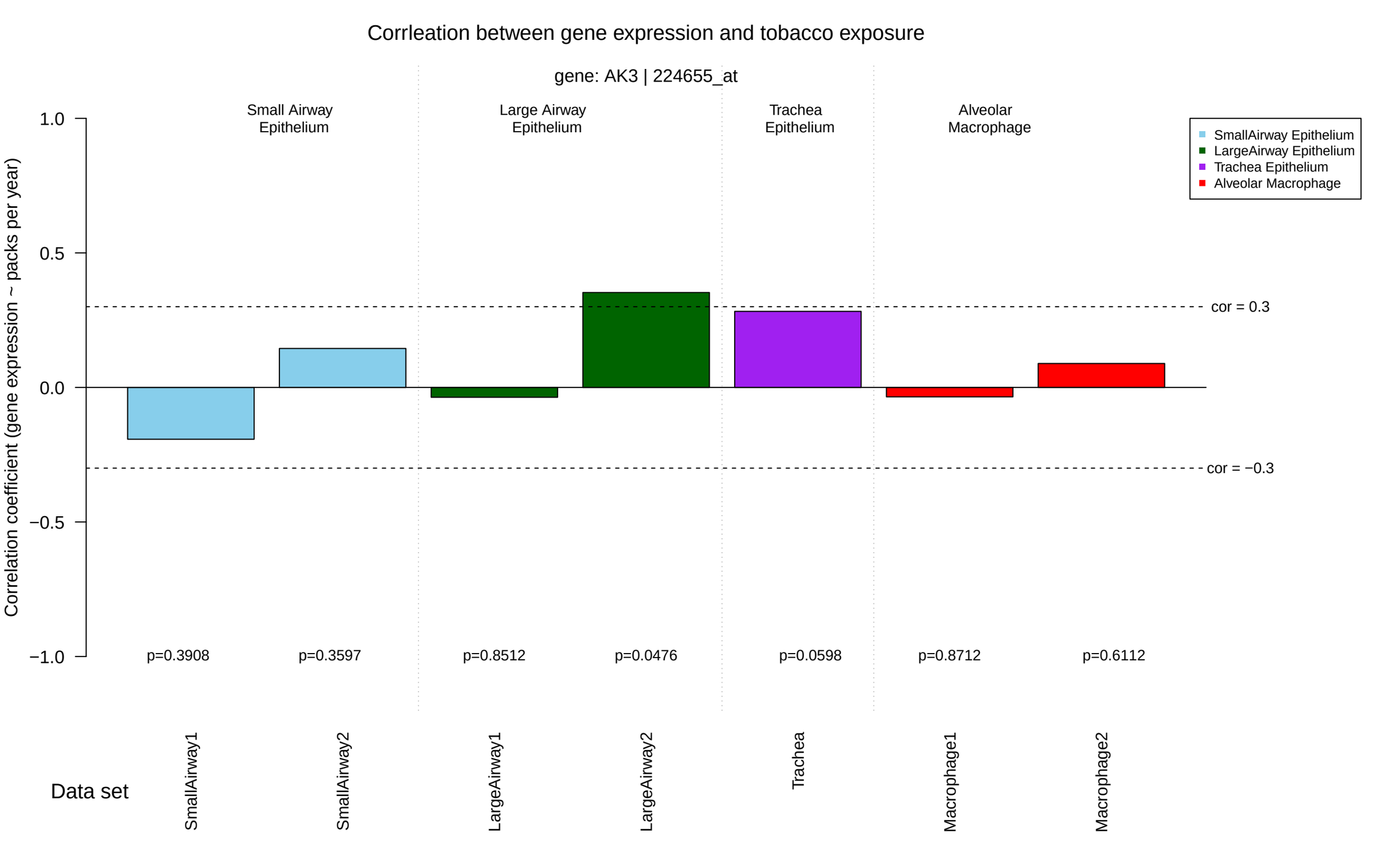

set: LargeAirway1 (p=0.1615)
AK3 | 224655_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1321)
AK3 | 224655_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=8e−04)
AK3 | 224655_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5083)
AK3 | 224655_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6629)
AK3 | 224655_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK3 | 224655_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3908
p=0.3597
p=0.8512
p=0.0476
p=0.0598
p=0.8712
p=0.6112
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2